Does Adding a Modality Really Make Positive Impacts in Incomplete Multi-Modal Brain Tumor Segmentation?

Yansheng Qiu, Kui Jiang, *Member, IEEE*, Hongdou Yao, Zheng Wang, *Senior Member, IEEE*, and Shin'ichi Satoh, *Senior Member, IEEE*

***Abstract*—Previous incomplete multi-modal brain tumor segmentation technologies, while effective in integrating diverse modalities, commonly deliver under-expected performance gains. The reason lies in that the new modality may cause confused predictions due to uncertain and inconsistent patterns and quality in some positions, where the direct fusion consequently raises the negative gain for the final decision. In this paper, considering the potentially negative impacts within a modality, we propose multi-modal Positive-Negative impact region Double Calibration pipeline, called PNDC, to mitigate misinformation transfer of modality fusion. Concretely, PNDC involves two elaborate pipelines, Reverse Audit and Forward Checksum. The former is to identify negative regions impacts of each modality. The latter calibrates whether the fusion prediction is reliable in these regions by integrating the positive impacts regions of each modality. Finally, the negative impacts region from each modality and miss-match reliable fusion predictions are utilized to enhance the learning of individual modalities and fusion process. It is noted that PNDC adopts the standard training strategy without specific architectural choices and does not introduce any learning parameters, and thus can be easily plugged into existing network training for incomplete multi-modal brain tumor segmentation. Extensive experiments confirm that our PNDC greatly alleviates the performance degradation of current state-of-the-art incomplete medical multi-modal methods, arising from overlooking the positive/negative impacts regions of the modality. The code is released at PNDC.**

***Index Terms*—Incomplete multi-modal, brain tumor segmentation, positive/negative impacts regions.**

I. Introduction

CLINICIANS employ various medical imaging modalities and protocols to obtain complementary diagnostic information. For example, different Magnetic Resonance Imaging (MRI) sequences—specifically, Fluid Attenuation Inversion Recovery (FLAIR), contrast enhanced T1-weighted (T1c), T1-weighted (T1), and T2-weighted (T2) are collectively utilized to intricately understand the spatial composition of brain tumors and their adjacent areas. In clinical practice, the occurrence of missing modalities is very common due to factors such as image corruption, artifacts, acquisition protocols, allergies to contrast agents, and cost constraints [1], [2], [3], [4], limiting the generality of the methods designed for complete modalities when confronted with missing modalities. Recently, the field of incomplete multi-modal segmentation has demonstrated remarkable advancements [4], [5], [6], [7], which is benefited from the fact that the modality reacts to the same object in different views. To facilitate modality fusion, most efforts have been developed to explore the informative components from different modalities [8], [9], [10], [11]. However, the performance of these modality-fusion-based methods is far below expectations, as depicted in Fig. 1 (a) and (b). This is because these methods default to the fact that the additional modalities inputs are all beneficial and do not focus on their negative impacts on fusion, which consequently declines the final prediction performance due to the fuzzy and confused knowledge fusion.

Additionally, taking the set of wrong predictions or unreliable predictions from all modalities as negative impacts regions is unreasonable. While the unreliable predictions in fusion outcomes may indicate erroneous predictions within the fusion itself, as illustrated in Fig.1 (c), the unreliable predictions from individual modalities serve only as explicit negative impact regions. This approach disregards the implicit negative impact regions, as individual modality errors and fused modality errors often intersect, as depicted in Fig.2 (a). Moreover, directly employing individual modality errors as negative contribution regions can lead to an excessive expansion of the area of interest, as seen in Fig. 1 (c) and (d). Regions where individual modalities yield unreliable predictions, along with their implicit negative impacts, are deemed as negative impacts regions on fusion prediction. The positive impacts regions of each modality are the opposite of the negative impacts regions.

That naturally arises a discussion whether the reliability of the extracted information by a specific modality should be re-investigated? To explore this issue further, two challenges

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Yansheng Qiu, Hongdou Yao, and Zheng Wang are with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: qiuyansheng@whu.edu.cn; yaohongdou0211@gmail.com; wangzwhu@whu.edu.cn).

Kui Jiang is with the School of Computer Science, Wuhan University, Wuhan 430072, China, and also with Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen 518000, China (e-mail: jiangkui@hit.edu.cn).

Shin'ichi Satoh is with the National Institute of Informatics, Tokyo 101-8430, Japan, and also with the Department of Information and Communication Engineering, Graduate School of Information Science and Technology, The University of Tokyo, Tokyo 113-8654, Japan (e-mail: satoh@nii.ac.jp).

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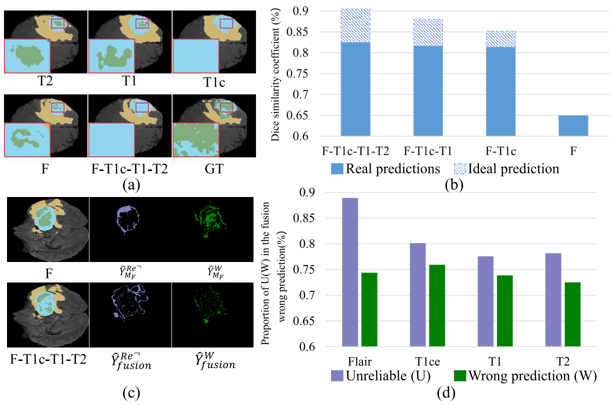

Fig. 1. Figure (a) displays the segmentation results of RFNet based on input of individual modality (T2, T1, T1c and F) and their fusion of four modalities (F-T1c-T1-T2). When integrating information from other modalities, wrong prediction of individual modality (like T1c) would deflect the prediction of fused modality from the label. Figure (b) showing obvious performance gap of the **real** and **ideal** prediction (A ideal prediction is considered to be correct if one of the modalities predictions is correct) accuracy for different combinations of modalities. It is speculated that the additional modality may provide the negative impacts to the final prediction due to the confused fusion. Figure (c) showcases the prediction (F and F-T1c-T1-T2), unreliable estimation ($Y_F^{Re^{\neg}}$ and $Y_{fusion}^{Re^{\neg}}$), and wrong prediction (Y_F^W and Y_{fusion}^W) of individual modality (F) and the fused modality (F-T1c-T1-T2). Figure (d) shows proportion of individual unreliable (fusion) prediction regions in the fusion wrong prediction. These results reveal two phenomenons: 1) the unreliable prediction regions are highly correlated to individual modality wrong redictions are difficult to correct; 2) the wrong redictions of individual modality may enlarge the scope of unreliable prediction of the fused modality.

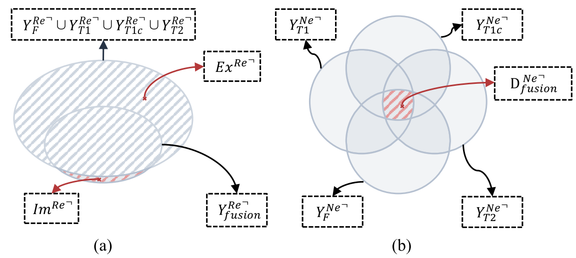

Fig. 2. Figure (a) exhibits the relationship of the unreliable prediction between the individual modality ($Y_F^{Re^{\neg}}$, $Y_{T1}^{Re^{\neg}}$, $Y_{T1c}^{Re^{\neg}}$, $Y_{T2}^{Re^{\neg}}$) and the modality fuison ($Y_{fusion}^{Re^{\neg}}$). Figure (b) reveals the positive information aliasing cross diverse modalities ($Y_F^{Ne^{\neg}}$, $Y_{T1}^{Ne^{\neg}}$, $Y_{T1c}^{Ne^{\neg}}$, $Y_{T2}^{Ne^{\neg}}$), where the intersection can be regarded as the desirable reliable prediction ($D_{fusion}^{Ne^{\neg}}$) in the fusion prediction.

become apparent. Then, the first challenge is that ***how to determine which positions of the modality have negative impacts on the fusion prediction.*** Suppose we have been given the reliable and unreliable regions of each modality prediction and fusion prediction. Some existing work [12], [13] utilizes Mixture of Experts (MoE) [14] to select compliant features from a set of feature combinations. Since we are concerned with pixel-level selection, MoE is not applicable to the challenge. Fortunately, we can first determine that the unreliable predictions for each modality are part of the negative impacts regions. Then, we need to find the implicit negative impacts regions of each modality and correspondingly combine them with unreliable predictions. However, the implicit negative impacts regions of all modalities are entangled outside the unreliable region and are difficult to quantify. The second challenge is that ***how to determine if reliable predictions of fusion are accurate.*** Assuming that we have obtained the corresponding positive impacts regions for each modality, it is logical that the fusion prediction should be reliable at these positions, as shown in Fig. 2(b). However, as shown in Fig. 1, there is still a discrepancy between the reliable fusion prediction and positive impacts combined by all modalities. These two critical challenges still need to be addressed.

To tackle these challenges, we propose a multi-modal **P**ositive-**N**egative impact region **D**ouble **C**alibration scheme to facilitate the incomplete multi-modal brain tumor segmentation, termed as PNDC. To begin with, we distinguish between unreliable and reliable predictions based on the entropy of individual modality predictions and the fused prediction. The fused unreliable predictions predominantly stem from the hybrid influence of the candidate modalities. Moreover, the overlap between the inaccurate predictions of individual modalities and those from their fusion is minimal. Directly treating the unreliable predictions of each modality as negative impact regions inevitably broadens the scope of these regions for each modality. Thus, we propose ***Reverse Audit*** to decouple the unreliable predictions of each modality, utilizing the unreliable regions in the fused predictions to determine the positive and negative impacts regions of each modality. Furthermore, we introduce ***Forward Checksum***, which utilizes the identified positive impact regions across all modalities to assess the accuracy of the reliable fused predictions. The detailed explanations will be given in Sec. III-B and Sec. III-C. The overarching aim of any multi-modal segmentation network is to minimize the negative impacts of each individual modality on the fused prediction and augment the region of fused reliable prediction. Finally, we intensify the learning process in the regions of fusion bias and negative impacts regions of each modality. Experiments on the BraTS 2020 dataset and BraTS 2018 demonstrate our PNDC can address the performance degradation observed in existing state-of-the-art unified models. It is noted that our proposed PNDC strategy is entirely pluggable and can be integrated into the existing incomplete Multi-modal technologies without introducing any learning parameters and model.

Overall, our contributions are threefold:

- We propose a pluggable multi-modal **P**ositive-**N**egative impact region **D**ouble **C**alibration scheme to facilitate the incomplete multi-modal brain tumor segmentation. To our knowledge this is the first attempt that considers the impact discrepancies of modality regions to facilitate segmentation.
- We investigate the real impacts of each modality on the specific regions and then harmonize the informative components of all modalities with reliability priors to verify the final prediction.
- Taking advantage of enhancing the learning of discovered fusion prediction bias regions and negative impacts regions for a single modality, PNDC could address the performance degradation observed in existing state-of-the-art incomplete medical multi-modal methods,

attributable to their oversight of the positive/negative impacts regions of the modality.

II. Related Work

A. Incomplete Multi-Modal Segmentation

Accurately segmenting brain tumors from multi-modal MRI images is useful for preoperative treatment planning. Various brain tumor segmentation methods were proposed [15], [16], [17]. However, in clinical practice, the occurrence of missing modalities is very common due to factors such as image corruption, artifacts, acquisition protocols, allergies to contrast agents, and cost constraints [1], [2], [3], limiting the generality of the methods designed for complete modalities when confronted with missing modalities. The incomplete multi-modal brain tumor segmentation task involves segmenting brain tumors from MRI images with various missing components [10], [18], [19], [20], [21]. Once one of the modalities is missing, the multi-modal fusion becomes unreachable, and the model may predict inaccurately. This task is more practical and challenging than standard brain tumor segmentation.

Generally, they can be grouped into three lines. The first category, modality generation-based methods [22], [23], aims to synthesize missing modalities and then leverages complete modalities as inputs to perform segmentation tasks. These methods are computationally cumbersome, requiring extra networks for synthesis. Since the optimization objective of generation and segmentation tasks are not coordinated, the synthesized data may affect segmentation performance. Dedicated segmentation based strategies as another line target to train a dedicated segmentation model for each possible combination of available modalities [24], [25]. Since there are fifteen possible combinations of modalities, these methods suffer from high training costs. The third line is based on multi-modal fusion approaches [5], [6], [7], [26], [27], which simply employ feature fusion strategies to aggregate features of all available sets of modalities to obtain fused representations for segmentation. Zhang et al. [6] have attempted to tackle this challenge by using transformers and CNN to build long-range dependencies within and across different modalities of MRI images for a modality-invariant representation. Ding et al. [5] proposed a Region-aware Fusion Module that aggregates multi-modal features of different regions adaptively to model the relationships between modalities and tumor regions. However, it is essential to note that such methods have used separate encoders for each modality and have focused on learning shared or invariant features. To address this problem, Zhao et al. [26] have recently proposed an adaptive feature interaction strategy based on the graph structure, which interacts with multi-modal features to accommodate multi-modal segmentation with different missing modalities.

Most of these studies concentrate on fusing multi-modal complementary, yet they often overlook the potential negative impacts of individual modalities on the fused prediction. In this paper, we refine the influence of each modality on the fusion process by disentangling the regions of negative and positive impact within the predictions of individual modalities, and we place greater emphasis on these regions during the training phase.

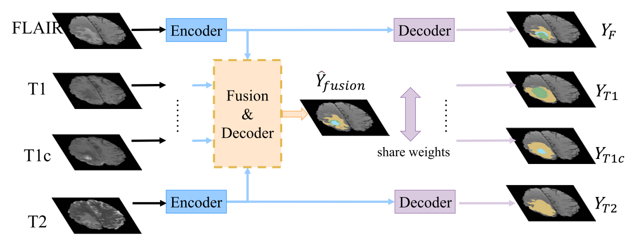

Fig. 3. Framework of current SOTA incomplete multi-modal brain tumor segmentation.

B. Uncertainty in Deep Learning

Uncertainty estimations [28], [29], [30] have been shown to be effective in detecting areas where networks are likely to make incorrect predictions in image analysis tasks. These estimations typically take the form of spatial uncertainty maps, visualizing a model's confidence in its predictions for each pixel in an image. A number of methods have been proposed to produce these maps, including Monte Carlo Dropout [31], Maximum Softmax Probability [32], and Deep Ensembles [33]. But the uncertainty method requires additional network branches to predict the distribution [34], then for each modal prediction and fusion prediction, adding uncertainty branches consumes computational resources. Another way to use conformal predictions [35], [36] is to determine the area of false negative. However, this type of approach works on the model inference phase, which requires traversing all the test samples, resulting in high computational cost and inability to optimize the network parameters.

In contrast to previous approaches, this paper focuses on both the positive and negative impacts of each modality on the fusion prediction, even identifying negative impacts in areas where confidence is high (reliable prediction). Without altering the network structure, we leverage the entropy of each pixel's probability distribution et al. [37], [38], [39] to distinguish between reliable and unreliable predictions for individual modalities and the fusion. We then utilize the fusion results to uncover regions of negative impact within the predictions of individual modalities and utilize the identified positive impact regions across all modalities to assess the accuracy of the reliable fused predictions.

III. Approach

A. Preliminaries

1) Incomplete Multi-Modal Brain Tumor Segmentation Baseline: Incomplete multi-modal brain tumor segmentation technologies, like RFNet [5], mmFormer [6] and A2FSeg [7] commonly predict and extract features for each modality, and then fuse the modality features to output fusion predictions, as is shown in Fig. 3.

2) Reliable and Unreliable Prediction: Following previously uncertainty work et al. [37], [38], [39], we utilize entropy of every pixel's probability distribution to filter unreliable and reliable prediction. Specifically, we denote $\mathbf{P}_{ij} \in \mathbb{R}^C$ as the softmax probabilities generated by the segmentation prediction head for individual modalities and fusion result at pixel ij, where C is the number of classes. Its entropy is computed by:

$$\mathcal{H}\left(\mathbf{p}_{ij}\right) = -\sum_{c=0}^{C-1} P_{ij}\left(c\right) \log P_{ij}\left(c\right), \tag{1}$$

where $P_{ij}(c)$ denotes the probability $\mathbf{P}_{ij}$ at the c-th dimension. Then, we define pixels whose entropy on top α_t as unreliable prediction at training epoch t. We define the prediction for the individual modalities and fusion at pixel ij as:

$$Y_{ij} = \begin{cases} Y^{Re}, & \text{if } \mathcal{H}\left(\mathbf{P}_{ij}\right) < \gamma_t, \\ Y^{Re^{\neg}}, & \text{otherwise}, \end{cases} \tag{2}$$

where γ_t represents the entropy threshold at t-th training step. We set γ_t as the quantile corresponding to α_t, *i.e.*, γ_t = `np.percentile(H.flatten(),100*(1)-`α_t`))`, where `H` is per-pixel entropy map. We adopt dynamic adjustment strategies to differentiate between unreliable and reliable prediction processes for better performance. Specifically, during the training procedure, the predictions tend to be reliable gradually. Based on this intuition, we adjust unreliable pixels' proportion α_t at every epoch with linear strategy, depicted as:

$$\alpha_t = \alpha_0 \cdot \left(1 - \frac{t}{\text{total epoch}}\right) \tag{3}$$

where α_0 is the initial proportion and is set to 20%, and t refers to the current training epoch.

3) Problem Definition: We temporarily consider the complete modalities condition comprising of $M = 4$ modalities, namely F, $T1$, $T1c$, and $T2$ (representing FLAIR, T1, T1c, and T2). The conditions of missing modalities are a subset of the currently considered condition. The presence of output branches for each modality in RFNet, mmFormer, and A2FSeg requires no modification. Thus we obtained the prediction of each modality (P_F, P_{T1}, P_{T1c}, and P_{T2}) with the fusion prediction (P_{fusion}) through the network. Then we proceed to the reliable (Y_F^{Re}, Y_{T1}^{Re}, Y_{T1c}^{Re}, Y_{T2}^{Re}, and Y_{fusion}^{Re}) and unreliable ($Y_F^{Re^{\neg}}$, $Y_{T1}^{Re^{\neg}}$, $Y_{T1c}^{Re^{\neg}}$, $Y_{T2}^{Re^{\neg}}$, and $Y_{fusion}^{Re^{\neg}}$) predictions from the individual modality and fusion predictions of the network through Eq. 2. As analyzed in Sec. I, the negative impacts region of each modality can be divided into implicit and explicit. The implicit negative impacts of all modalities are entangled outside the unreliable region. In Sec. III-B, we would use the unreliable fusion predictions to decouple the positive impacts and negative impacts regions from the unreliable predictions of each modality. Considering the existence of inconsistencies in the positive impacts region for each modality and the reliable prediction of the fusion prediction, in Sec. III-C, we employ the ascertained positive impacts regions of all modalities to differentiate the accuracy of the fused reliable predictions. Table I details the notations and their responding explanations.

B. Reverse Audit

As demonstrated in Sec. I, we investigate the negative impacts of individual modality on the final fusion prediction.

TABLE I
SUMMARY OF MAIN NOTATIONS

Notation	Description	Notation	Description
P	Segmentation prediction	$\mathbf{G}$	Ground truth
Y^{Re}	Reliable fusion prediction	$Y^{Re^{\neg}}$	Unreliable fusion prediction
Y^{Ne}	Negative modal prediction	$Y^{Ne^{\neg}}$	Positive modal prediction
Ex^{Ne}	Explicit negative prediction	Im^{Ne}	Implicit negative prediction
$D_{fusion}^{Ne^{\neg}}$	Desirable reliable prediction	Y_{fusion}^{Bi}	Fusion bias
Mu^{Ne}	Mutual negative prediction	Y^{W}	Wrong prediction

It is observed that the unreliable region of the fusion prediction is a subset of the negative regions of all modalities, so we **assume** that:

$$Y_{fusion}^{Re^{\neg}} = \bigcup_{\hat{M} \in M} Y_{\hat{M}}^{Ne}, \tag{4}$$

where M involves $\{F, T1, T1c, T2\}$, and Y_F^{Ne}, Y_{T1}^{Ne}, Y_{T1c}^{Ne}, Y_{T2}^{Ne}, Y_{fusion}^{Ne} are the regions of negative impacts of individual modality to the fusion prediction, respectively.

In addition, we consider unreliable predictions of individual modality as explicit negative impact regions since these regions provide perceptible probabilistic perturbations to fusion, as shown in Ex^{Ne} (explicit negative impact) regions in Fig. 4. The implicit negative impacts regions denote the mixed distribution of positive and negative results of all modalities and are difficult to determine by quantitative means. The only clue is that the combination of explicit negative impact regions and implicit negative impact regions contributes to the unreliable prediction of fusion. So, unreliable prediction regions that belong to the fusion but not to a single modality are considered to be implicit negative impacts from other modalities, as shown in Mu^{Ne} (mutual negative impact) regions in Fig. 4.

Specifically, as is shown in Fig. 4 (a), for a single modality *e.g.,* F, implicit negative region of mutual modalities decomposed from its unreliable prediction is the region of mixed action belonging to $T1$, $T1c$ and $T2$, called mutual negative prediction Mu^{Ne}. And the other regions are explicit negative impacts regions Ex^{Ne}:

$$Y_{\hat{M}}^{Re^{\neg}} = \begin{cases} Mu^{Ne}, & \text{where } Y \in Y_{fusion}^{Re^{\neg}} \text{ and } Y \notin Y_{\hat{M}}^{Re^{\neg}}, \\ Ex^{Ne}, & \text{otherwise}, \end{cases} \tag{5}$$

Then the region of implicit negative impacts of the F exists in Mu_{T1}^{Ne}, Mu_{T1c}^{Ne}, and Mu_{T2}^{Ne}. Since the mutual negative impacts region is independent of the current modality, then the region they share is implicit negative impacts region of the F, as is shown in Fig. 4 (b):

$$Im_{\hat{M}^i}^{Ne} = \bigcup_{\hat{M} \in M \text{ and } \hat{M} \neq \hat{M}^i} Mu_{\hat{M}}^{Ne}. \tag{6}$$

The implicit negative impacts regions and the explicit negative impacts regions together constitute the negative impacts

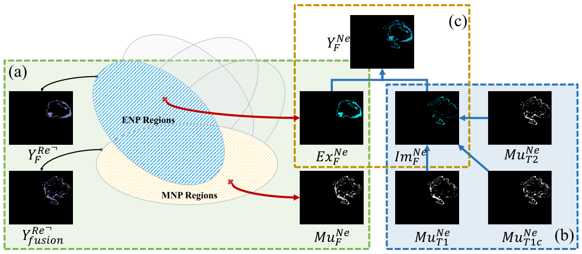

Fig. 4. Pipeline of the reverse audit. (a) We use the unreliable prediction of fusion to decouple individual modality explicit negative impacts regions and mutual negative impacts regions from the unreliable prediction of individual modalities. (b) Extracting implicit negative impacts regions from mutual negative impacts regions. (c) Implicit negative impacts regions combine with explicit negative impacts regions to get each modality negative impacts regions. Positive impacts regions are the opposite of negative impacts.

regions of the current modality, as is shown in Fig. 4 (c):

$$Y_{\hat{M}}^{Ne} = Im_{\hat{M}}^{Ne} \cup Ex_{\hat{M}}^{Ne}. \tag{7}$$

Then the opposite of the negative impacts regions are positive impacts regions $Y_{\hat{M}}^{Ne\neg}$.

C. Forward Checksum

Ideally, the fusion prediction of a multi-modal network should involve the positive impact regions of all modalities. However, in practical scenarios, the network design and data distribution cause discrepancies in the prediction information, i.e., all modalities are positively impact regions but the fused prediction is unreliable. To address this, we calibrate the fusion predictions using the positive impact regions for each modality extracted.

Specifically, at the same position, each modality contributes positively to the modality fusion, and we believe that the fusion predictions deserve to be reliable predictions in this position:

$$D_{fusion}^{Ne\neg} = \bigcap_{\hat{M}\in M} Y_{\hat{M}}^{Ne\neg}. \tag{8}$$

Unreliable predictions of fusion at these positions are recognized as fusion bias:

$$Y_{fusion}^{Bi}, \text{ where } Y \in D_{fusion}^{Ne\neg} \text{ and } Y \notin Y_{fusion}^{Re}. \tag{9}$$

D. Enhanced Learning of Ambiguous Regions

At this point, we have obtained the negative impact regions for each modality $Y_{\hat{M}}^{Ne}$, and again determined the fusion bias Y_{fusion}^{Bi} due to the model fusion based on the positive impact regions for all modalities. In network learning, we want these regions to be as small as possible. For each modality, from the previous analysis we get that small cross-entropy regions tend to contribute positively, so the most direct way is to improve the prediction accuracy of these regions. Similarly, we would like the network to pay more attention to the fusion prediction bias introduced by the model.

The segmentation task loss is to train each pixel with its ground-truth label using cross-entropy:

$$L_{seg} = CE\left(\sigma\left(\mathbf{P}\right), \mathbf{G}\right). \tag{10}$$

Here, CE denotes the cross-entropy loss, σ denotes the softmax function and $\mathbf{G}$ denotes the ground-truth label. We make the negative impacts of each modality act on the region of the ground-truth label to do additional constraints:

$$L_m = CE\left(\sigma\left(\mathbf{P_M}\right) * Y_M^{Ne}, \mathbf{G} * Y_M^{Ne}\right). \tag{11}$$

We perform the same for the fusion bias regions brought about by the model:

$$L_{Bi} = CE\left(\sigma\left(\mathbf{P_{fusion}}\right) * Y_{fusion}^{Bi}, \mathbf{G} * Y_{fusion}^{Bi}\right), \tag{12}$$

In PNDC, it consists of a Reverse Audit Stage and a forward Checksum-Stage in order to address the issue of missing modalities for incomplete brain tumor segmentation. Therefore, the overall objective function of our PNDC is:

$$L = L_{seg} + \lambda_1 * L_m + \lambda_2 * L_{Bi}. \tag{13}$$

where λ_1 and λ_2 are trade-off factors to balance the impacts of each term. L_{seg} consists of standard dice loss and multi-class cross-entropy loss for accurate segmentation task.

Algorithm 1 PNDC Algorithm

Input: P_F, P_{T1}, P_{T1c}, and P_{T2}: Individual modality prediction with size of $N \times H \times W$; P_{fusion}: Fusion prediction with size of $N \times H \times W$;

Output: negative impacts regions of the each modality: Y_F^{Ne}, Y_{T1}^{Ne}, Y_{T1c}^{Ne}, and Y_{T2}^{Ne}; Fusion bias Y_{fusion}^{Bi};

for $i = 1$; $i < Epoch$; $i++$ **do**
  Get all unreliable prediction regions $Y_F^{Re\neg}$, $Y_{T1}^{Re\neg}$, $Y_{T1c}^{Re\neg}$, $Y_{T2}^{Re\neg}$, $Y_{fusion}^{Re\neg}$ based on Eq.2;
  Get all reliable prediction regions Y_F^{Re}, Y_{T1}^{Re}, Y_{T1c}^{Re}, Y_{T2}^{Re}, Y_{fusion}^{Re} based on Eq.2;
  Get all individual modality mutual negative impacts regions Mu_F^{Ne}, Mu_{T1}^{Ne}, Mu_{T1c}^{Ne}, Mu_{T2}^{Ne}, based on Eq.5;
  Get all individual modality explicit negative impacts regions Ex_F^{Ne}, Ex_{T1}^{Ne}, Ex_{T1c}^{Ne}, Ex_{T2}^{Ne}, based on Eq.5;
  Get all individual modality implicit negative impacts regions Im_F^{Ne}, Im_{T1}^{Ne}, Im_{T1c}^{Ne}, Im_{T2}^{Ne}, based on Eq.6;
  Get all individual modality negative impacts regions Y_F^{Ne}, Y_{T1}^{Ne}, Y_{T1c}^{Ne}, Y_{T2}^{Ne}, based on Eq.7;
  Invert the individual modality negative impacts regions to get the individual modality positive impacts regions $Y_F^{Ne\neg}$, $Y_{T1}^{Ne\neg}$, $Y_{T1c}^{Ne\neg}$, $Y_{T2}^{Ne\neg}$;
  Get all individual modality consistent and desirable reliable regions $D_{fusion}^{Ne\neg}$ based on Eq.8;
  Get fusion bias Y_{fusion}^{Bi} based on Eq.9;
end for

IV. Experiments

A. Datasets and Evaluation Metrics

We use three different datasets from the Multi-modal Brain Tumor Segmentation Challenge (BraTS) [40]: BraTs2023, BraTs2020 and BraTs2018. The subjects in the BraTs2020 and BraTs2018 datasets contain four distinct MRI modalities, *i.e.*, Flair, T1c, T1 and T2, and The subjects in the

BraTs2023 datasets contain a) native (T1) and b) post-contrast T1-weighted (T1Gd), c) T2-weighted (T2), and d) T2 Fluid Attenuated Inversion Recovery (T2-FLAIR). We split BraTs2020 and BraTs2018 datasets into train set, validate set, and test set, respectively, in the same scheme as [5], [41], [42], and [43], and we split BraTs2023 into 7:1:2 for training, validation, and test, respectively. Since not all brain tumors contain enhancing areas, we apply a post-processing strategy to reduce false positives for enhancing tumor regions. Specifically, if the number of pixels predicted as enhancing tumor is below a certain threshold (e.g., fewer than 500 pixels), we classify this as a false positive and reassign these pixels as non-enhancing tumor regions, as is in [5], [42], and [44]. Dice coefficient [45] is used to measure the segmentation performance of the proposed method. The whole tumor, tumor core, and enhancing tumor are composed of BG, NCR/NE, ED, and ET combinations [6]. $Dice_{\bar{k}}$ denotes the Dice score of the tumor class $\bar{k}$. Larger Dice scores represent that predictions are more similar to the ground truth and thus indicate better segmentation accuracy. We also evaluated all methods via the 95th percentile of the Hausdorff distance (HD95) and sensitivity for performance quantification. Higher sensitivity and smaller HD95 indicate better segmentation accuracy.

B. Implementation Details

We deployment PNDC to current representative work: RFNet, mmFormer, and A2FSeg. Considering the differences between the different methods, customized training details are used.

1) RFNet: For image pre-processing, the MRI images are skull-stripped, co-registered, and re-sampled to $1mm^3$ resolution by the data collector. Following [43], we cut out the black background area outside the brain and normalize each MRI modality to zero mean and unit variance in the brain region. During training, input images are randomly cropped to $80 \times 80 \times 80$ and are then augmented with random rotations, intensity shifts, and mirror flipping. We train our network for 600 epochs with a batch size of 2. Adam [46] is leveraged to optimize the network with β_1 and β_2 of 0.9 and 0.999, respectively, and the weight decay is set to $1e^{-5}$.

2) mmFormer: We start by segmenting $128 \times 128 \times 128$ patches that slide over the test images, with 50% overlaps between neighboring patches, and batch size is 2. The final segmentation map is obtained by fusing the predictions of these patches. Random flip, crop, and intensity shifts are employed for data augmentation. The network is trained with the Adam optimizer with an initial learning rate of $2e^{-4}$ for 1000 epochs.

3) A2FSeg: The MRI images went through a sequence of preprocessing steps, including co-registration to the same anatomical template, resampling to the same resolution ($1mm^3$), and skull stripping. We use the Adam optimizer [46], with an initial learning rate of 0.01. Due to the limitation of GPU memory, each volume is randomly cropped into multiple patches with the size of $128 \times 128 \times 128$ for training. The network is trained for 400 epochs. In the inference stage, we use a sliding window to produce the final segmentation prediction of the input image.

C. Performance Comparison

We conduct comprehensive experiments on various cases with missing MRI modalities using the BraTS2018, BraTS2020, and BraTS2023 datasets.

As illustrated in Table II, our proposed method significantly improves the average scores for the whole tumor, the tumor core, and the enhancing tumor of A2FSeg, RFNet, and mmFormer in $Dice$ and achieves a new SOTA for the incomplete modalities. Notably, PNDC improves the mmFormer with an average dice score increase of 0.48%, 0.62%, 1.00%, improves RFNet with an average dice score increase of 0.98%, 1.32%, 3.14%, and improves A2FSeg with an average dice score increase of 0.46%, 0.78%, 0.65% on the whole tumor, the tumor core, and the enhancing tumor regions. Notably, our PNDC performs better than RFNet in all cases with missing modalities, which demonstrates the effectiveness of the modality positive&negative impacts awareness. Additionally, our method surpasses mmFormer and A2FSeg in the three key indicators across the majority of the 15 multi-modal combinations. Specifically, it secures the leading position in 39 out of 45 subcategories compared to mmFormer, and in 41 out of 45 subcategories compared to A2FSeg.

Further, validation on the BraTS2018 dataset further highlights the efficacy of our approach. As illustrated in Table III, our approach significantly elevates the average $Dice$ scores for the whole tumor, tumor core, and enhancing tumor regions in A2FSeg, RFNet, and mmFormer, achieving a new SOTA for incomplete modalities. PNDC improves mmFormer by 0.39%, 0.65%, and 1.81%, RFNet by 0.76%, 0.81%, and 4.25%, and A2FSeg by 0.49%, 0.66%, and 0.51% in these regions, respectively. Moreover, our method demonstrates superiority over mmFormer, RFNet, and A2FSeg across most of the 15 multi-modal combinations, taking the lead in 43 out of 45 subcategories compared to mmFormer, 44 out of 45 compared to RFNet, and 37 out of 45 compared to A2FSeg.

Lastly, validation was conducted on the BraTS2023 dataset to confirm the robustness of our model. As illustrated in Table IV, PNDC consistently improves the performance of A2FSeg, RFNet, and mmFormer for the whole tumor, tumor core, and enhancing tumor regions, establishing a new SOTA for incomplete modalities. Specifically, PNDC improves mmFormer by 0.36%, 0.06%, and 0.28%, RFNet by 1.09%, 1.80%, and 2.37%, and A2FSeg by 0.52%, 0.18%, and 0.33% in the respective regions. In addition, our method outperforms mmFormer, RFNet, and A2FSeg in the three indicators of the vast majority of 15 multi-modal combinations. Specifically, we achieved the lead place in 35 out of 45 subcategories compared to mmFormer, achieved the lead place in 44 out of 45 subcategories compared to RFNet, and achieved the lead place in 36 out of 45 subcategories compared to A2FSeg.

Moreover, we observe that our proposed method effectively alleviates the issue of diminishing performance growth rates as the number of modalities increases. Specifically, in Table II, the accuracy gaps from from T2 modality to T1c and T2 modality to T1, T1c, and T2 modality to FLAIR, T1, T1c, and T2 of mmFormer are optimized by PNDC, improving from 1.67%, 0.35%, and 2.48% to 1.87%, 0.54%, and 2.70%.

TABLE II

RESULTS OF STATE-OF-THE-ART UNIFIED MODELS (MMFORMER, RFNET, A2FSEG, U-HVED, ROBUSTSEG) AND THE PNDC (* MEANS DEPLOYED ON THEIR BASIS)FOR DIVERSE MODALITY COMBINATIONS ON THE BRATS 2020 DATASET. DICE SIMILARITY COEFFICIENT (DSC) [%] IS EMPLOYED FOR EVALUATION.COMPLETE, CORE AND ENHANCING DENOTE THE DICE SCORES OF THE WHOLE TUMOR, THE TUMOR CORE AND THE ENHANCING TUMOR, RESPECTIVELY. THE BEST AND SECOND BEST SCORES COMPARED TO BASELINE ARE BOLD AND UNDERLINED, RESPECTIVELY. MODALITIES PRESENT ARE DENOTED BY •, WHILE THE MISSING ONES BY ○

Modalities	FLAIR	○	○	○	•	○	○	•	○	•	•	•	•	•	○	•	Average
	T1	○	○	•	○	○	•	•	•	○	○	•	•	○	•	•	
	T1c	○	•	○	○	•	•	○	○	○	•	•	○	•	•	•	
	T2	•	○	○	○	•	○	○	•	•	○	○	•	•	•	•	
Complete	U-HVED	80.75	68.54	54.93	82.69	83.37	71.58	85.01	81.58	87.40	86.13	87.10	88.07	88.33	84.27	88.81	81.24
	RobustSeg	82.20	71.39	71.41	82.87	85.97	76.84	88.10	85.53	88.09	87.33	88.87	89.24	88.68	86.63	89.47	84.17
	mmFormer	**86.14**	78.47	77.97	87.40	87.81	81.75	89.79	**87.81**	89.85	89.34	90.11	90.57	90.38	88.16	90.64	87.08
	mmFormer*	86.05	**79.40**	**79.30**	**87.70**	**87.92**	**82.58**	**89.92**	87.73	**90.21**	**90.31**	**90.54**	**90.71**	**91.01**	**88.46**	**91.16**	**87.53**
	RFNet	86.05	76.77	77.16	87.32	87.74	81.12	89.73	87.73	89.87	89.89	90.69	90.60	90.68	88.25	91.11	86.98
	RFNet*	**86.68**	**79.48**	**80.29**	**87.84**	**88.74**	**83.11**	**90.26**	**88.43**	**90.32**	**90.45**	**91.03**	**90.86**	**91.23**	**89.17**	**91.51**	**87.96**
	A2FSeg	86.97	80.57	79.22	88.79	**88.78**	82.54	89.74	87.72	90.43	90.12	90.62	91.02	91.23	**89.01**	**91.50**	87.88
	A2FSeg*	**87.86**	**82.28**	**80.72**	**89.09**	88.64	**84.11**	**89.79**	**88.42**	**90.74**	**90.30**	**90.65**	90.91	**91.28**	88.88	91.41	**88.34**
Core	U-HVED	57.43	73.01	36.73	51.15	77.85	76.49	55.10	85.53	61.87	76.86	79.51	63.46	78.68	79.99	80.40	67.19
	RobustSeg	61.88	76.88	54.30	60.72	82.44	80.28	68.18	66.46	68.20	81.85	82.76	70.46	81.89	82.85	82.87	73.45
	mmFormer	70.86	83.99	65.92	68.79	84.75	83.56	73.84	**73.41**	74.61	84.78	85.22	75.58	85.32	84.17	84.61	78.69
	mmFormer*	**71.30**	**84.70**	**67.28**	**70.58**	**85.60**	**85.26**	**74.11**	72.95	**75.28**	**85.27**	**85.46**	**75.70**	**85.68**	**85.24**	**85.30**	**79.31**
	RFNet	71.02	81.51	66.39	69.19	83.45	83.40	73.07	73.13	74.14	84.65	85.07	75.19	84.97	83.47	85.21	78.23
	RFNet*	**71.43**	**83.58**	**67.44**	**71.71**	**85.47**	**84.21**	**74.89**	**73.56**	**75.71**	**85.67**	**85.41**	**76.59**	**86.07**	**85.65**	**85.92**	**79.55**
	A2FSeg	74.02	84.71	71.19	72.29	86.38	85.58	76.17	75.92	76.46	85.86	86.94	78.08	86.75	87.03	87.34	80.98
	A2FSeg*	**75.69**	**86.48**	**71.37**	**74.33**	**87.01**	**86.70**	**76.82**	**76.45**	**77.82**	**86.85**	**87.40**	**78.28**	**87.38**	**87.15**	**87.49**	**81.76**
Enhancing	U-HVED	28.70	66.59	12.33	20.87	68.74	67.82	22.53	28.73	30.48	69.53	71.32	30.60	69.84	69.74	70.50	48.55
	RobustSeg	36.46	67.91	28.99	34.68	71.42	70.11	39.67	39.92	42.19	70.78	71.77	43.90	71.17	71.87	71.52	55.49
	mmFormer	46.34	80.08	38.03	**42.37**	79.39	80.07	**45.92**	46.76	48.64	**81.89**	82.09	50.29	78.73	79.29	79.92	64.08
	mmFormer*	**49.12**	**80.09**	**41.48**	40.82	**80.24**	**81.19**	44.59	**50.73**	**50.69**	81.47	**82.27**	**51.66**	**81.06**	**80.06**	**80.69**	**65.08**
	RFNet	46.29	74.85	37.30	38.15	75.93	78.01	40.98	45.65	49.32	76.67	76.81	49.92	77.12	76.99	78.00	61.47
	RFNet*	**49.93**	**75.81**	**39.18**	**44.86**	**78.12**	**79.55**	**48.43**	**50.43**	**51.81**	**78.46**	**81.46**	**52.00**	**78.73**	**80.18**	**80.29**	**64.61**
	A2FSeg	48.86	75.94	41.68	46.67	76.81	75.69	47.80	49.17	50.45	76.07	75.96	50.93	76.19	**76.63**	76.02	62.99
	A2FSeg*	**50.34**	**75.95**	**43.30**	**47.93**	**76.98**	**76.85**	**47.52**	**49.78**	**51.84**	**76.11**	**76.95**	**51.77**	**76.21**	76.15	**76.95**	**63.64**

Table VI reports that our PNDC also improves the baselines regarding sensitivity and Hausdorff distance (95%) on BraTS2020 in most cases. Table VII shows that we executed paired t-tests on the Brats2020 dataset. It is evident that both the complete and core paired t-tests yield a P-value less than 0.05, signifying the statistically significant enhancements achieved by our method. Furthermore, we conduct a comparative analysis of RFNet and PNDC using the validation dataset of BraTs2023 through the official online evaluation server. The average scores for all missing cases are presented in Table V. Notably, PNDC demonstrates significant performance improvements, highlighting its potential for further enhancement.

Fig. 5 illustrates that our method can segment brain tumors well in various missing scenarios. Compared to mmFormer, RFNet, and A2FSeg, PNDC can significantly improve segmentation accuracy and accurately distinguish between NCR/NET and other categories, which indicates that PNDC can assist models in capturing better spatial context. Besides, PNDC adopts the standard training strategy without specific architectural choices and thus can be easily plugged into existing incomplete multi-modal brain tumor segmentation to address performance degradation arising from overlooking the positive/negative regions of modalities.

D. Ablation Study

The following experiments study the impacts of the different components of the proposed approach, and the evaluation is done on the BraTs2020 dataset. For our initial attempts at PNDC, we use RFNet [5] as the baseline.

1) Reverse Audit/Forward Checksum: Our objective is to enable the network to find and focus on negative impact regions for each modality and mismatched unreliable fusion predictions by designing two elaborate pipelines, Reverse Audit, and Forward Checksum. As depicted in the top part of Table IX, the PNDC surpasses the baseline after Reverse Audit and Forward Checksum. This demonstrates that both Reverse Audit and Forward Checksum are effective in accomplishing the design goals. Furthermore, while Forward Checksum can lead to some improvement, it is not as effective as Reverse Audit. This is because individual modalities are the basis of fusion, and without focusing on the negatively impact regions of individual modalities, mismatched unreliable fusion predictions are more difficult to discover. Our approach, PNDC, employs both Forward Checksum and Reverse Audit, thereby achieving superior accuracy compared to the baseline and other strategies that focus on individual modalities predictions or fusion predictions. This is due to the fact that PNDC effectively addresses the challenge of handling negative impacts from individual modalities to fusion and mismatched bias due to model design during the feature extraction and modal fusion stages.

2) Compare With Uncertainty: To confirm our hypothesis: at some locations, there are regions of negative impacts of modal pairs fusion, but each modal unreliable prediction fails to respond to these regions of negative impacts. We change the

TABLE III

RESULTS OF STATE-OF-THE-ART UNIFIED MODELS (MMFORMER, RFNET, A2FSEG, U-HVED, ROBUSTSEG) AND THE THE PNDC (* MEANS DEPLOYED ON THEIR BASIS)FOR DIVERSE MODALITY COMBINATIONS ON THE BRATS 2018 DATASET. DICE SIMILARITY COEFFICIENT (DSC) [%] IS EMPLOYED FOR EVALUATION.COMPLETE, CORE AND ENHANCING DENOTE THE DICE SCORES OF THE WHOLE TUMOR, THE TUMOR CORE AND THE ENHANCING TUMOR, RESPECTIVELY. THE BEST AND SECOND BEST SCORES COMPARED TO BASELINE ARE BOLD AND UNDERLINED, RESPECTIVELY. MODALITIES PRESENT ARE DENOTED BY •, WHILE THE MISSING ONES BY ◦

Modalities	FLAIR	◦	◦	◦	•	◦	◦	•	◦	•	•	•	•	•	◦	•	Average
	T1	◦	◦	•	◦	◦	•	•	•	◦	◦	•	•	◦	•	•	
	T1c	◦	•	◦	◦	•	•	◦	◦	◦	•	•	◦	•	•	•	
	T2	•	◦	◦	◦	•	◦	◦	•	•	◦	◦	•	•	•	•	
Complete	U-HVED	80.90	62.40	52.40	82.10	82.70	66.80	84.20	82.20	87.50	85.50	86.20	88.00	88.60	83.30	88.80	80.10
	RobustSeg	82.24	73.31	70.11	85.69	85.19	77.18	88.24	84.78	88.28	88.51	88.73	88.81	89.27	86.01	89.45	84.39
	mmFormer	85.64	77.11	77.01	87.17	86.87	80.52	88.85	86.92	89.48	89.51	89.66	89.76	90.11	87.07	90.14	86.38
	mmFormer*	**85.84**	**78.50**	**77.85**	**87.33**	**86.89**	**81.39**	**89.11**	**86.98**	**89.83**	**89.73**	**89.88**	**90.02**	**90.51**	**87.16**	**90.47**	**86.77**
	RFNet	84.30	74.93	74.68	86.46	86.39	78.59	88.78	86.15	89.12	89.17	89.71	89.68	90.06	**86.78**	90.26	85.67
	RFNet*	**86.21**	**76.89**	**76.89**	**87.41**	**87.45**	**80.52**	**89.28**	**87.18**	**89.81**	**89.59**	**89.84**	**90.18**	**90.63**	83.91	**90.64**	**86.43**
	A2FSeg	87.17	79.87	80.27	88.28	87.56	82.40	**89.34**	87.37	89.90	89.77	90.03	90.25	**90.73**	87.80	**90.84**	87.43
	A2FSeg*	**87.65**	**81.15**	**80.85**	**88.94**	**87.66**	**83.71**	89.30	**87.93**	**90.33**	**90.23**	**90.44**	**90.53**	90.69	**88.55**	90.83	**87.92**
Core	U-HVED	54.10	66.70	37.20	50.40	73.70	69.70	55.30	57.20	59.70	72.90	74.20	61.50	75.60	75.30	76.40	64.00
	RobustSeg	57.49	76.83	47.90	53.57	80.20	78.72	60.68	62.19	61.16	80.62	81.06	64.38	80.72	80.33	80.86	69.78
	mmFormer	68.04	80.68	64.61	64.53	82.06	82.03	**70.77**	70.42	**70.44**	81.98	82.79	72.01	82.05	82.45	82.36	75.82
	mmFormer*	**68.20**	**82.07**	**65.01**	**64.44**	**83.30**	**83.28**	70.74	**70.99**	70.34	**82.85**	**83.73**	**72.59**	**82.86**	**83.34**	**83.32**	**76.47**
	RFNet	67.62	80.99	64.42	64.89	83.27	82.22	71.59	70.89	70.82	82.94	83.77	73.09	83.54	83.97	84.02	76.53
	RFNet*	**68.31**	**81.52**	**65.36**	**66.19**	**83.93**	**82.40**	**72.73**	**72.37**	**71.90**	**83.68**	**84.26**	**74.14**	**84.57**	**84.00**	**84.69**	**77.34**
	A2FSeg	71.31	83.37	69.93	69.21	84.20	83.65	73.28	73.58	72.55	84.04	84.33	74.70	84.52	84.28	84.54	78.49
	A2FSeg*	**71.96**	**84.95**	**70.18**	**71.52**	**85.82**	**85.30**	**73.98**	**74.37**	**73.89**	**86.26**	**84.84**	**75.45**	**85.15**	**84.43**	**84.69**	**79.15**
Enhancing	U-HVED	30.80	65.50	13.70	24.80	70.20	67.00	24.20	30.70	34.60	70.30	71.10	34.10	71.20	71.10	71.70	50.00
	RobustSeg	28.97	67.07	17.29	25.69	69.71	69.06	32.13	32.01	33.84	70.30	70.78	36.41	70.88	70.10	71.13	51.02
	mmFormer	40.32	76.02	32.74	35.62	75.82	75.63	39.36	42.45	43.46	76.33	76.69	44.00	75.84	76.18	76.43	59.12
	mmFormer*	**42.07**	**77.18**	**39.14**	**37.11**	**77.79**	**77.23**	**40.68**	**43.71**	**44.94**	**77.80**	**77.24**	**45.09**	**77.89**	**78.17**	**77.88**	**60.93**
	RFNet	40.71	69.43	34.43	33.92	73.01	70.73	39.68	41.42	43.77	72.84	73.17	44.79	73.13	72.56	73.21	57.12
	RFNet*	**43.71**	**76.32**	**38.10**	**39.54**	**76.66**	**76.28**	**43.63**	**45.25**	**46.70**	**75.92**	**77.85**	**49.35**	**77.47**	**76.86**	**76.86**	**61.37**
	A2FSeg	43.64	**72.32**	41.76	40.68	**72.94**	72.12	**43.03**	44.31	43.65	72.46	72.51	44.36	**73.06**	**72.61**	72.84	58.81
	A2FSeg*	**44.02**	72.30	**43.94**	**42.89**	72.93	**72.52**	42.70	**44.92**	**44.85**	**72.76**	**72.85**	**44.83**	72.84	71.91	**73.44**	**59.32**

TABLE IV

RESULTS OF STATE-OF-THE-ART UNIFIED MODELS (MMFORMER, RFNET, A2FSEG) AND THE THE PNDC (* MEANS DEPLOYED ON THEIR BASIS)FOR DIVERSE MODALITY COMBINATIONS ON THE BRATS 2023 DATASET. DICE SIMILARITY COEFFICIENT (DSC) [%] IS EMPLOYED FOR EVALUATION.COMPLETE, CORE AND ENHANCING DENOTE THE DICE SCORES OF THE WHOLE TUMOR, THE TUMOR CORE AND THE ENHANCING TUMOR, RESPECTIVELY. THE BEST AND SECOND BEST SCORES COMPARED TO BASELINE ARE BOLD AND UNDERLINED, RESPECTIVELY. MODALITIES PRESENT ARE DENOTED BY •, WHILE THE MISSING ONES BY ◦

Modalities	T2-FLAIR	◦	◦	◦	•	◦	◦	•	◦	•	•	•	•	•	◦	•	Average
	T1	◦	◦	•	◦	◦	•	•	•	◦	◦	•	•	◦	•	•	
	T1Gd	◦	•	◦	◦	•	•	◦	◦	◦	•	•	◦	•	•	•	
	T2	•	◦	◦	◦	•	◦	◦	•	•	◦	◦	•	•	•	•	
Complete	mmFormer	**87.97**	81.07	80.21	90.55	89.77	83.92	91.62	89.20	92.00	92.62	92.79	92.21	93.07	90.13	93.11	89.34
	mmFormer*	87.94	**81.51**	**81.07**	**91.39**	**89.93**	**84.29**	**92.15**	**89.23**	**92.34**	**92.93**	**93.07**	**92.55**	**93.43**	**90.23**	**93.48**	**89.70**
	RFNet	86.95	80.19	80.18	89.69	89.47	83.10	90.86	88.91	91.07	91.43	91.98	91.70	92.15	89.80	92.36	88.65
	RFNet*	**88.16**	**82.67**	**82.25**	**90.55**	**90.28**	**84.97**	**91.68**	**89.88**	**91.78**	**92.42**	**92.83**	**92.30**	**92.81**	**90.70**	**92.92**	**89.74**
	A2FSeg	88.61	82.77	83.22	91.85	89.64	85.23	92.37	89.31	92.49	92.75	93.10	92.72	93.15	89.96	93.27	90.02
	A2FSeg*	**89.65**	**83.83**	**84.08**	**91.96**	**90.54**	**85.90**	**92.57**	**90.22**	**92.81**	**92.69**	**93.11**	**93.09**	**93.29**	**90.93**	**93.45**	**90.54**
Core	mmFormer	73.32	88.17	**72.54**	75.95	89.39	**89.57**	78.21	**76.77**	78.73	90.63	**91.23**	79.49	90.84	90.28	90.92	83.73
	mmFormer*	**74.50**	**88.50**	70.23	**76.30**	**89.86**	89.12	**79.04**	76.20	**79.13**	**90.79**	91.18	**79.64**	**90.99**	**90.31**	**91.12**	**83.79**
	RFNet	73.39	88.79	72.36	74.19	91.18	90.06	77.46	77.33	77.78	90.43	91.14	78.99	91.31	91.53	91.60	83.83
	RFNet*	**77.24**	**90.64**	**74.50**	**77.25**	**92.30**	**91.35**	**79.56**	**79.34**	**80.11**	**92.00**	**92.30**	**81.44**	**92.19**	**91.97**	**92.33**	**85.63**
	A2FSeg	77.42	**91.43**	76.83	78.12	**92.27**	**92.13**	79.88	79.14	79.84	**92.06**	**92.65**	80.80	92.16	92.55	92.05	85.95
	A2FSeg*	**78.31**	91.29	**77.04**	**78.28**	91.74	91.85	**80.52**	**79.57**	**80.49**	91.95	92.51	**81.21**	**92.44**	**92.56**	**92.20**	**86.13**
Enhancing	mmFormer	56.71	84.59	**54.17**	56.29	85.89	86.06	61.27	**61.11**	62.82	**86.60**	**86.86**	64.21	86.29	**86.72**	86.72	73.75
	mmFormer*	**58.27**	**85.01**	53.11	**56.86**	**86.13**	**86.19**	**62.22**	60.45	**63.57**	86.29	86.70	**64.74**	**86.99**	86.59	**87.28**	**74.03**
	RFNet	56.83	84.95	52.50	**55.99**	86.78	86.22	60.41	60.82	62.56	86.64	87.08	64.22	86.91	87.05	87.07	73.73
	RFNet*	**61.66**	**87.28**	**55.96**	58.66	**87.86**	**88.51**	**62.84**	**64.41**	**66.13**	**88.48**	**88.40**	**67.40**	**87.75**	**88.29**	**88.02**	**76.10**
	A2FSeg	60.71	86.85	**59.92**	61.61	**87.55**	**87.48**	63.89	62.78	63.96	87.32	**87.95**	64.97	87.80	87.81	88.00	75.90
	A2FSeg*	**61.39**	**86.86**	59.80	**62.21**	87.50	87.47	**64.69**	**63.68**	**65.08**	**87.44**	87.60	**65.78**	**87.87**	**87.90**	**88.21**	**76.23**

output of Forward Checksum and Reverse Audit to unreliable predictions for each modality and fused unreliable predictions.

As depicted in the Table IX, the results of our PNDC significantly outperform those using unreliable predictions

Fig. 5. Visualization of the predicted segmentation maps in BraTs2020. NCR/NET, ED and ET are illustrated in green, yellow and blue, respectively. Left top: four image modalities. Others: segmentation maps predicted by mmFormer, RFNet, A2FSeg, and our PNDC deployment derived from these models from all fifteen combinations of image modalities and the corresponding ground truth.

directly, demonstrating the effectiveness of focusing on each modal negative impacts region and fusing prediction biases in improving model prediction accuracy. The reason that the direct use of unreliable predictions is lower than the baseline

TABLE V
COMPARISONS USING THE OFFICIAL ONLINE EVALUATION UNDER AVERAGE HAUSDORFF95 AND DICE SCORES OF FIFTEEN MULTI-MODAL COMBINATIONS WITH THE RFNET THE PNDC (* MEANS DEPLOYED ON THEIR BASIS) ON BRATS2023. 'COM', 'COR' AND 'ENH' DENOTES THE WHOLE TUMOR, THE TUMOR COREAND THE ENHANCING TUMOR. THE RESULTS COMPARED TO BASELINE WITH UNDERLINE DENOTE THE SECOND BEST PERFORMANCE AND WITH BOLD SHOWS THE BEST PERFORMANCE

Methods	Dice(%)↑			Hausdorff95(mm)↓		
	Com	Cor	Enh	Com	Cor	Enh
RFNet	86.83	77.53	64.73	13.19	16.84	35.37
RFNet*	**88.33**	**79.51**	**67.11**	**9.73**	**15.29**	**33.48**

TABLE VI
COMPARISONS UNDER AVERAGE HAUSDORFF95 AND SENSITIVITY SCORES OF FIFTEEN MULTI-MODAL COMBINATIONS WITH THE U-HVED, ROBUSTSEG, MMFORMER RFNET, A2FSEG, AND THE PNDC (* MEANS DEPLOYED ON THEIR BASIS) ON BRATS2020. 'COM', 'COR' AND 'ENH' DENOTES THE WHOLE TUMOR, THE TUMOR COREAND THE ENHANCING TUMOR. THE RESULTS COMPARED TO BASELINE WITH UNDERLINE DENOTE THE SECOND BEST PERFORMANCE AND WITH BOLD SHOWS THE BEST PERFORMANCE

Methods	sensitivity(%)↑			Hausdorff95(mm)↓		
	Com	Cor	Enh	Com	Cor	Enh
U-HVED	99.60	99.78	99.78	8.63	11.99	58.38
RobustSeg	99.48	99.78	99.77	5.75	9.85	57.30
mmFormer	99.60	99.81	99.82	2.90	6.21	44.64
mmFormer*	**99.61**	**99.83**	**99.83**	**2.41**	**4.67**	**40.42**
RFNet	99.60	**99.84**	99.82	3.72	9.24	51.88
RFNet*	**99.66**	99.82	**99.84**	**3.17**	**6.18**	**45.40**
A2FSeg	**99.91**	99.93	99.92	**6.13**	6.95	49.46
A2FSeg*	99.90	**99.94**	**99.93**	6.16	**6.30**	**48.58**

TABLE VII
THE P-VALUE WITH PAIRED T-TEST OF PNDC WITH MMFORMER, RFNET, AND A2FSEG

Compared Method	Complete	Core	Enhancing
mmFormer	4.47e-03	5.42e-05	1.02e-01
RFNet	1.47e-03	9.31e-02	9.42e-02
AF2Seg	2.12e-02	2.46e-02	5.03e-01

TABLE VIII
THE INFLUENCE ON THE ACCURACY OF TRADE-OFF FACTORS λ_1 AND λ_2

λ_1	T_M	Dice(%)		
		Complete	Core	Enhancing
1.0	1.0	87.52	78.62	64.38
0.5	0.5	87.63	78.76	64.55
0.4	0.6	87.50	79.13	64.92
0.6	0.4	87.50	78.67	64.33
0.2	0.8	**87.96**	**79.55**	**64.61**
0.8	0.2	87.46	78.73	64.38

is that the negative contributing regions of the modes are more neglected.

3) Effect of Trade-off Factors: In Table VIII we present the impact of selecting different trade-off factors, λ_1 and λ_2, on the final results. The results reach optimal levels when λ_1 is set to 0.2 and λ_2 to 0.8. Notably, when λ_2 is assigned a lower value than λ_1, model performance declines significantly. This reduction in performance arises because λ_1 primarily affects the model's intermediate outputs, whereas λ_2, which carries greater importance, directly influences the final output.

TABLE IX
PROPOSED COMPONENTS AND VARIANTS OF THE PNDC ON COMPARISON OF SEGMENTATION EFFECTS COMPARING TO BASELINE ON BRATS2020 DATASET

Operation	Dice(%)		
	Complete	Core	Enhancing
Baseline	86.98	78.23	61.47
+Forward Checksum	87.16	78.38	62.25
+Reverse Audit	87.09	78.47	63.73
Uncertainty	87.15	78.28	61.26
Incorrect predictions	87.29	78.08	62.44
PNDC	**87.96**	**79.55**	**64.61**

TABLE X
PROPOSED PNDC ON COMPARISON OF TRAINING COAST COMPARING TO BASELINE ON BRATS2020 DATASET

Method	Train time cost (second per image)	Inference time cost (second per image)	GPU memory usage of training (GB)
RFNet	0.8765	0.0884	22.90
RFNet*	0.8955	0.0884	23.06

4) Incorrect Predictions: In order to verify that uncertainty is more responsive to the negative impacts of individual modalities than mis-prediction, we use incorrect predictions (IPDC) instead of unreliable predictions in PNDC. As depicted in the Table IX, the results of our PNDC significantly outperform IPDC, demonstrating the effectiveness of PNDC with uncertainty.

5) Times Comparison: PNDC is based only on the existing SOTA training framework and does not introduce new models and parameters without multiplication or convolution operation, so GFLOPs cannot be calculated. In addition, PNDC is only used in the training phase, not in the inference phase. As shown in Table X, we record the training cost for PNDC compared to the baseline on the same computational platform, observing that we hardly affect the training time.

V. DISCUSSIONS

As illustrated in Fig. 6, we display the uncertainty prediction maps for each modality, the negative impact regions identified by PNDC, as well as the predictions generated by both RFNet and PNDC. Our findings reveal that the proposed PNDC effectively concentrates on the regions where each modality negatively affects fusion, performing more precisely in this regard than the uncertainty estimations of individual modalities. Additionally, as shown in the last three rows, there are instances where PNDC successfully detects regions where each modality contributes negatively to fusion yet struggles to rectify erroneous predictions within these areas. This limitation may arise because, in certain cases, the fused prediction's uncertainty maps lack sufficient sensitivity to address such erroneous predictions. Furthermore, given that we apply a uniform threshold across uncertainty maps for both fused and individual modality predictions, this may also contribute to the occurrence of false predictions. In future work, this issue could be addressed by using distinct uncertainty thresholds for fused and individual modality predictions.

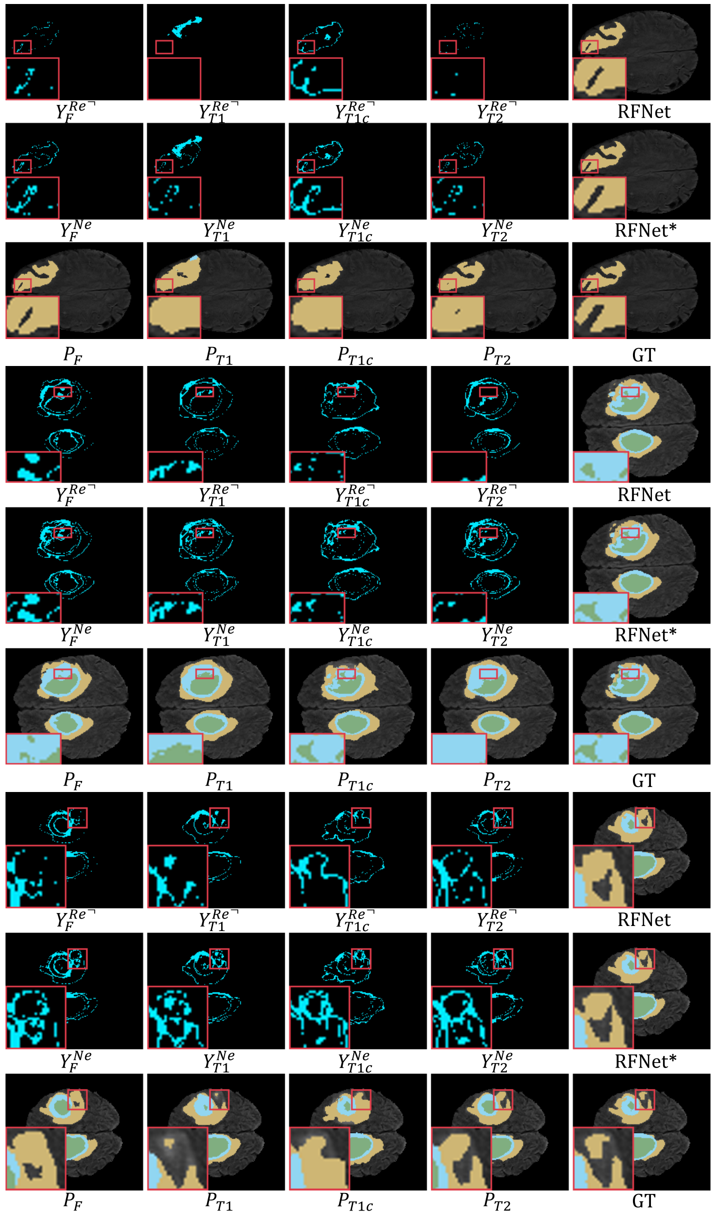

Fig. 6. Visualization of the Uncertainty maps for each modality, regions of negative impact found by PNDC, predictions of RFNet and predictions of PNDC (*).

VI. Conclusion

In this work, we propose an inimitable pipeline focusing on discovering negative impacts regions of individual modality for incomplete modality brain tumor segmentation to alleviate the performance degradation caused by the fusion of negative information without introducing any learning parameters and model. As part of our process, we use uncertainty to distinguish between reliable and unreliable regions in the prediction of each modality and in the prediction of the fusion. Next, we introduce the reverse audit to locate positive/negative impact regions of each modality using predictive unreliability predictions. Then, We utilize the positive influence region in the forward checksum to identify the fusion prediction bias introduced by the model design. Extensive experiments demonstrate that PNDC can address the performance degradation of state-of-the-art segmentation framework arising from overlooking the positive/negative impacts regions of the modality on the widely used BraTs2023, BraTs2020, and BraTs2018 benchmarks.

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